

Kalyan Kumar Kurusali

Email: kalyan.k@careernb.com | Contact: +1 (940) 315-7539 | Location: Texas, USA | <https://www.linkedin.com/in/kkk1999/>

Summary

Experienced Data Scientist with nearly 5 years of practical experience in banking and IT infrastructure, specializing in Python, R, SQL, Hadoop, Spark, Power BI, and Tableau. Proven expertise in generating actionable insights and improving efficiency through machine learning, fraud detection, data visualization, and big data analytics. Built and maintained ETL pipelines supporting fraud detection, credit risk modeling and regulatory reporting. Developed interactive dashboards in Power BI and Tableau for real-time banking analytics, enhancing compliance tracking and customer experience. Leveraged real-time data monitoring tools to build forecasting models for maintenance planning, automated SQL reporting, and uptime, cost, and performance predictions. Designed and deployed scalable, cloud-native data pipelines and analytics frameworks across AWS (S3, Lambda, Redshift, SageMaker), Azure (Data Factory, Synapse, Azure ML), and GCP (BigQuery, Vertex AI, Dataflow) to support real-time processing, automation, and business intelligence. Delivered strategic, cloud-integrated data solutions aligned with operational, regulatory, and business goals.

Skills

- Programming Languages:** SQL, Python, R, Scala
- Database:** MySQL, PostgreSQL, MongoDB, Snowflake, NoSQL
- Tools:** R Studio, Jupyter Notebook, Tableau, Power BI, Hadoop, Spark, Airflow, Talend, Excel Solver, Docker, R Shiny, Git
- Library:** Pandas, Numpy, Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, LightGBM, Matplotlib, Seaborn, NLTK, SpaCy
- Machine Learning:** Decision Trees (CART), Ridge/Lasso Regression, ResNet, Naïve Bayes, KNN, SVM, Boosting, Random Forest, Clustering, NLP, Time-Series Forecasting
- Cloud:** S3, Redshift, Glue, Lambda, EMR, SageMaker, EC2, IAM, RDS, Athena, CloudFormation, Azure Data Factory, Azure Synapse Analytics, Azure Machine Learning, Azure Blob Storage, Azure SQL Database, Azure Kubernetes Service (AKS), Big Query, Vertex AI, Cloud Functions, Dataflow, Cloud Storage
- Methodologies:** Agile, Scrum, SDLC, Waterfall
- Operating Systems:** Windows, MS Server, Linux, MacOS
- Others:** MS Office (Word, Excel, PowerPoint), Zoom, Mentoring, CI/CD, Docker, REST APIs

Experience

Northern Trust, AL | [Data Scientist](#) Jan 2024 - Current

- Developed and deployed scalable machine learning models on **AWS SageMaker** and **Azure Machine Learning**, predicting loan defaults, customer churn and credit risk with improved accuracy.
- Engineered real-time data pipelines using **AWS Glue**, **S3**, and **Lambda** for processing streaming data from banking systems, enabling proactive fraud detection and compliance alerts.
- Designed **Tableau** and **R Shiny** dashboards for interactive visualizations of fraud metrics, credit exposure, and operational risk, streamlining executive decision-making.
- Leveraged **GCP BigQuery** for querying terabyte-scale datasets and used **Vertex AI** for building and testing ML models, accelerating model deployment by 40%.
- Applied Natural Language Processing (NLP) techniques to analyse transaction descriptions and detect suspicious activities, reducing fraud investigation time by 50%.
- Conducted model interpretability and bias analysis to meet regulatory standards, collaborating closely with cross-functional teams (finance, risk, compliance) under Agile methodology.
- Automated data reporting workflows using **Azure Synapse Analytics** and **Data Factory**, improving reporting turnaround time and reducing manual effort.

Accenture, India | [Data Scientist II](#) Mar 2021 – Aug 2022

- Built predictive models using Random Forests, KNN, Naïve Bayes, and Logistic Regression to support marketing strategy, lead scoring, and risk analysis for financial clients.
- Automated ML pipeline orchestration using **Python**, **SQL**, and **Azure Data Factory**, reducing manual tasks and enabling continuous deployment.
- Extracted, cleaned, and analysed structured and unstructured data stored in **MySQL**, **MongoDB**, and **PostgreSQL**, integrating results into dynamic dashboards with **Power BI**.
- Used **GCP Dataflow** for large-scale ETL and deployed models to **Vertex AI**, enabling near real-time predictions across multiple client environments.
- Collaborated with client stakeholders to transform business needs into analytical solutions, providing detailed documentation and knowledge transfer sessions.
- Enhanced data ingestion workflows using **Azure Blob Storage** and **Azure SQL**, ensuring high availability and scalability of data-driven applications.

Accenture, India | [Jr Data Scientist](#) Jan 2019 – Feb 2021

- Developed supervised learning models (Decision Trees, Naïve Bayes) and clustering models using K-Means to segment user profiles and enhance targeting strategies.
- Built robust ML pipelines using **Scikit-learn**, **TensorFlow**, and **Docker**, containerizing models and deploying them via **AWS ECS** and **Azure Kubernetes Service (AKS)**.
- Performed complex data preprocessing, feature engineering, and time-series modeling using Python (Pandas, NumPy), SQL, and **GCP BigQuery**.
- Designed automated reporting systems in **Power BI Service**, integrating with **Azure SQL Database** and **AWS RDS** for real-time analytics.
- Applied clustering and segmentation techniques to identify customer patterns, increasing personalization and campaign effectiveness by 25%.
- Developed scenario models in Excel using the Analytic Solver Platform and supported business teams with statistical forecasting and risk simulations.
- Used **Jupyter Notebook** and **R Studio** for exploratory data analysis, modeling, and documentation, facilitating smooth project handovers and stakeholder engagement.

Education

Master of Science in Advanced Data Analytics | GPA 3.9/4.0

University of North Texas, Denton, Texas

Honors: Dean's Excellence Scholarship

Certification

- Machine Learning with Cart in R
- Introduction to Regression in R
- Building Web Applications with Shiny in R
- Learning Excel: Data Analysis
- SQL for Data Analysis
- Oracle Cloud Infrastructure 2024 Generative AI
- Introduction to Prompt Engineering for Generative AI